1 Experiments

In subsections 1.1 and 1.2, the models did not see the other appendix type column only the column associated to the experiment.

1.1 experiment 1

This is nasal appendix only with appendix duration. Model stayed the same. The appendix labels were taken out after splitting into two columns with binary representation. The nasal appendix was determined by a 1 if the appendix label column was either “n” or “n+c” and a 0 otherwise.

1.2 experiment 2

This is creak appendix only with appendix duration. Model stayed the same. The appendix labels were taken out after splitting into two columns with binary representation. The creak appendix was determined by a 1 if the appendix label column was either “c” or “n+c” and a 0 otherwise.

1.3 experiment 3

This was with both appendix type but removing the appendix duration to test what label was the model looking at more. The model had access to both appendix type binary columns.

2 Results

2.1 Nasal only

2.1.1 Metrics

Table 1: Classification Report for Nasal Appendix Model

type	**precision**	**recall**	**f1-score**	**support**
0 (clear)	70%	71%	70%	211
1 (normal)	68%	66%	67%	192
accuracy			69%	403
macro avg	69%	69%	69%	403
weighted avg	69%	69%	69%	403

Table 1 is the metric calculations for the model that only saw the nasal appendix type. Table 2 is the validation accuracy and the out-of-box scores. The model has not performed better from looking at both appendix types.

Table 2: Other Report for Nasal Appendix Model

type	**result**
Validation accuracy	68.73%
OOB score	74.12%

2.1.2 Graphs

In figure 1 is the imporatance for the nasal appendix. We can see that the appendix_dur and vowel duration has decreased in amplitude but the actual tag was not useful.

Figure 1: Feature importances using MDI for the nasal appendix model.

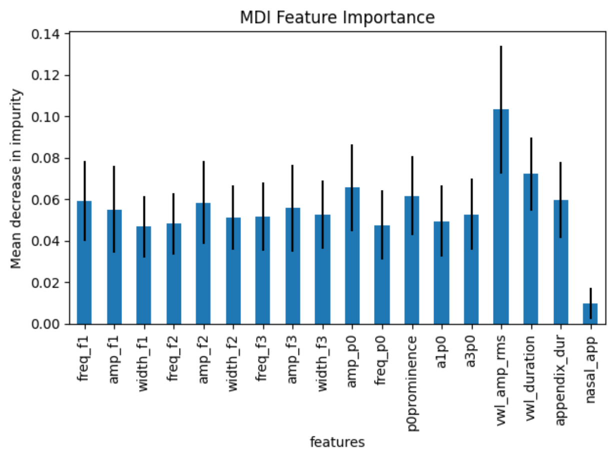
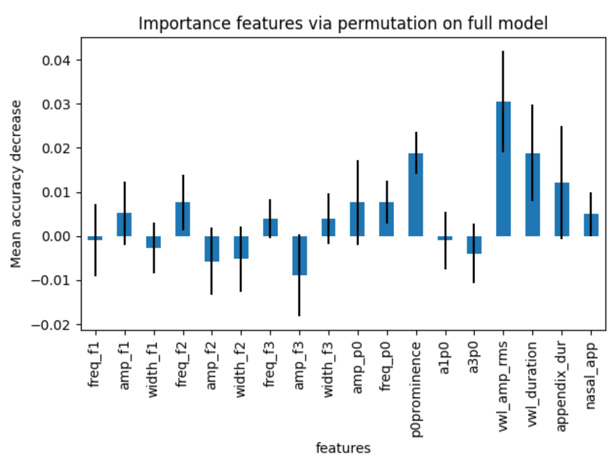


Figure 2: Feature permutation using MDI for the nasal appendix model.



In figure 2, we see that the vowel amplitude rms is still the feature. The appendix and vowel duration has decreased. It is interesting that for both figures, the appendix tag did not have much of an impact.

2.2 Creak only

2.2.1 Metrics

Table 3: Classification Report for Creak Appendix Model

type	precision	recall	f1-score	support
0 (clear)	72%	70%	71%	211
1 (normal)	68%	70%	69%	192
accuracy			70%	403
macro avg	70%	70%	70%	403
weighted avg	70%	70%	70%	403

Table 4: Other Report for Creak Appendix Model

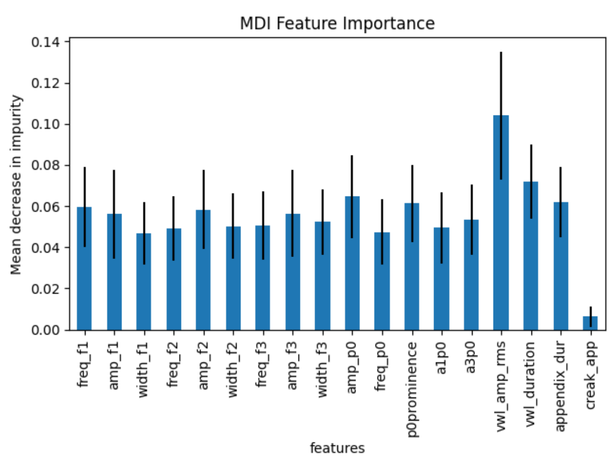
type	result
Validation accuracy	69.98%
OOB score	74.67%

Table 3 is the metric calculations for the model that only saw the creak appendix type. Table 4 is the validation accuracy and the out-of-box scores. The model has not performed better from looking at both appendix types.

2.2.2 Graphs

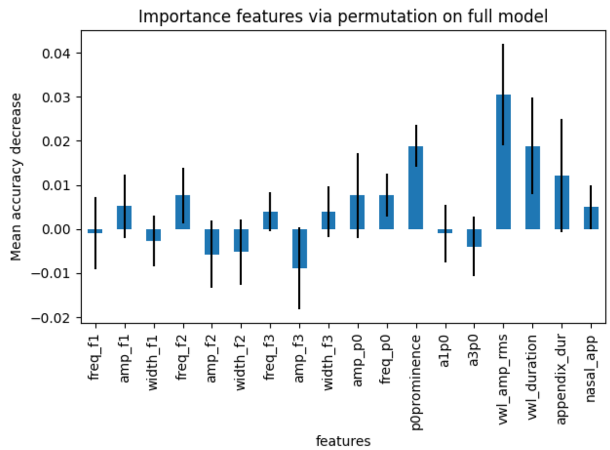
In figure 1 is the imporatance for the nasal appendix. We can see that the appendix_dur and vowel duration has decreased in amplitude but the actual tag was not useful.

Figure 3: Feature importances using MDI for the creak appendix model.



In figure 2, we see that the vowel amplitude rms is still the feature. The appendix and vowel duration has decreased. It is interesting that for both figures, the appendix tag did not have much of an impact.

Figure 4: Feature permutation using MDI for the creak appendix model.



2.3 Both

2.3.1 Metrics

Table 5: Classification Report for Creak Appendix Model

type	precision	recall	f1-score	support
0 (clear)	72%	73%	72%	211
1 (normal)	69%	69%	69%	192
accuracy			71%	403
macro avg	71%	71%	71%	403
weighted avg	71%	71%	71%	403

Table 6: Other Report for Creak Appendix Model

type	result
Validation accuracy	70.72%
OOB score	72.75%

Table 5 is the metric calculations for the model that only saw the creak appendix type. Table 6 is the validation accuracy and the out-of-box scores. The model has not performed better from looking at both appendix types without looking at the appendix duration.

2.3.2 Graphs

The vowel duration rms is still the main feature that the model looks at. but looking at figure 6 that the nasal appendix has played some role when it comes to the permutation.

Figure 5: Feature importances using MDI.

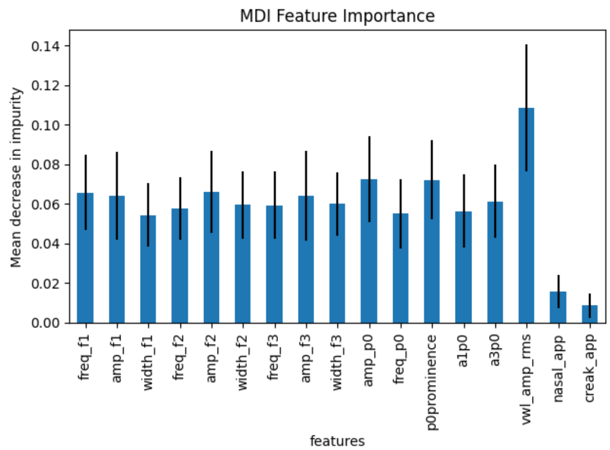


Figure 6: Feature permutation using MDI.